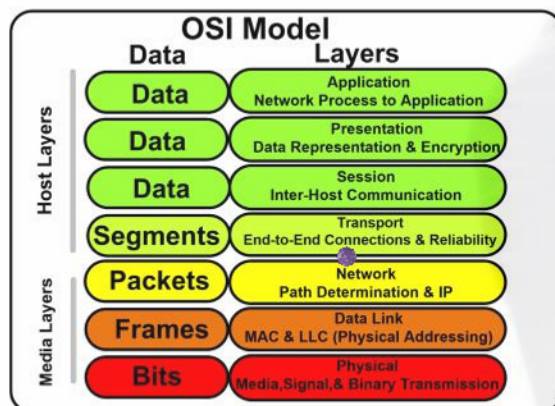


# DAY 5 OF 75 DAYS ETHICAL HACKING COURSE

The **OSI model** (or **Open Systems Interconnection Model**) is an absolute fundamental model used in networking. This critical model provides a framework dictating how all networked devices will send, receive and interpret data.

One of the main benefits of the OSI model is that devices can have different functions and designs on a network while communicating with other devices. Data sent across a network that follows the uniformity of the OSI model can be understood by other devices.

The OSI model consists of seven layers which are illustrated in the diagram below. Each layer has a different set of responsibilities and is arranged from Layer 7 to Layer 1. At every individual layer that data travels through, specific processes take place, and pieces of information are added to this data, which is what we'll come to discuss in the upcoming tasks within this room. However, for now, we only need to understand that this process is called encapsulation and what the OSI model looks like in the diagram below:



## APPLICATION

The application layer of the OSI model is the layer that you will be most familiar with. This familiarity is because the application layer is the layer in which protocols and rules are in place to determine how the user should interact with data sent or received.

Everyday applications such as email clients, browsers, or file server browsing software such as FileZilla provide a friendly, **Graphical User Interface (GUI)** for users to interact with data sent or received. Other protocols include **DNS (Domain Name System)**, which is how website addresses are translated into IP addresses.

What is the technical term that is given to the name of the software that users interact with?

graphical user interface

## Layer 6 - Presentation

Layer 6 of the OSI model is the layer in which standardisation starts to take place. Because software developers can develop any software such as an email client differently, the data still needs to be handled in the same way — no matter how the software works.

This layer acts as a translator for data to and from the application layer (layer 7). The receiving computer will also understand data sent to a computer in one format destined for in another format. For example, when you send an email, the other user may have another email client to you, but the contents of the email will still need to display the same.

Security features such as data encryption (like HTTPS when visiting a secure site) occur at this layer.

What is the main purpose that this Layer acts as?

translator

## Layer 5 - Session

Once data has been correctly translated or formatted from the presentation layer (layer 6), the session layer (layer 5) will begin to create a connection to the other computer that the data is destined for. When a connection is established, a session is created. Whilst this connection is active, so is the session.

The session layer (layer 5) synchronises the two computers to ensure that they are on the same page before data is sent and received. Once these checks are in place, the session layer will begin to divide up the data sent into smaller chunks of data and begin to send these chunks (packets) one at a time. This dividing up is beneficial because if the connection is lost, only the chunks that weren't yet sent will have to be sent again — not the entire piece of the data (think of it as loading a save file in a video game).

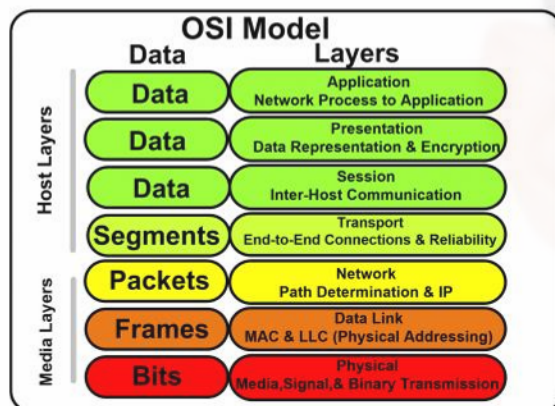
What is worthy of noting is that sessions are unique — meaning that data cannot travel over different sessions, but in fact, only across each session instead



What is the technical term for "small chunks of data"?

packets

## Layer 4 - Transport



Layer 4 of the OSI model plays a vital part in transmitting data across a network and can be a little bit difficult to grasp. When data is sent between devices, it follows one of two different protocols that are decided based upon several factors:

- TCP
- UDP

Let's begin with TCP. The **T**ransmission **C**ontrol **P**rotocol (**TCP**). Potentially hinted by the name, this protocol is designed with reliability and guarantee in mind. This protocol

reserves a constant connection between the two devices for the amount of time it takes for the data to be sent and received.  
Not only this, but TCP incorporates error checking into its design. Error checking is how TCP can guarantee that data sent from the small chunks in the session layer (layer 5) has then been received and reassembled in the same order.

Let's summarise the advantages and disadvantages of TCP in the table below:

#### Advantages of TCP

Guarantees the accuracy of data.

Capable of synchronising two devices to prevent each other from being flooded with data.

Performs a lot more processes for reliability.

#### Disadvantages of TCP

Requires a reliable connection between the two devices. If one small chunk of data is not received, then the entire chunk of data cannot be used

TCP is significantly slower than UDP because more work has to be done by the devices using this protocol.

TCP is used for situations such as file sharing, internet browsing or sending an email. This usage is because these services require the data to be accurate and complete (no good having half a file!).

Now let's move onto the **User Datagram Protocol** (or **UDP** for short). This protocol is not nearly as advanced as its brother - the TCP protocol. It doesn't boast the many features offered by TCP, such as error checking and reliability. In fact, any data that gets sent via UDP is sent to the computer whether it gets there or not. There is no synchronisation between the two devices or guarantee; just hope for the best, and fingers crossed.

Whilst this sounds disadvantageous, it does have its merits, which we'll layout in the table below:

#### Advantages of UDP

UDP is much faster than TCP.

UDP leaves the application layer (user software) to decide if there is any control over how quickly packets are sent.

UDP does not reserve a continuous connection on a device as TCP does.

#### Disadvantages of UDP

UDP doesn't care if the data is received.

It is quite flexible to software developers in this sense.

This means that unstable connections result in a terrible experience for the user.

Using the same example as before, we can now see that only Packets #1 and #3 have been received by the "Computer", meaning that half of the image is missing. UDP is useful in situations where there are small pieces of data being sent. For example, protocols used for discovering devices (ARP and DHCP) that we discussed

### Answer the questions below

What is the name of this Layer?

transport

What does TCP stand for?

transmission control protocol

What does UDP stand for?

user datagram protocol

What protocol guarantees the accuracy of data?

tcp

What protocol doesn't care if data is received or not by the other device?

udp

## Layer 3 - Network

The third layer of the OSI model (network layer) is where the magic of routing & re-assembly of data takes place (from these small chunks to the larger chunk). Firstly, routing simply determines the most optimal path in which these chunks of data should be sent. Whilst some protocols at this layer determine exactly what is the "optimal" path that data should take to reach a device, we should only know about their existence at this stage of the networking module. Briefly, these protocols include **OSPF** (Open Shortest Path First) and **RIP** (Routing Information Protocol). The factors that decide what route is taken is decided by the following:

- What path is the shortest? I.e. has the least amount of devices that the packet needs to travel across.
- What path is the most reliable? I.e. have packets been lost on that path before?
- Which path has the faster physical connection? I.e. is one path using a copper connection (slower) or a fibre (considerably faster)?

At this layer, everything is dealt with via IP addresses such as 192.168.1.100. Devices such as routers capable of delivering packets using IP addresses are known as Layer 3 devices — because they are capable of working at the third layer of the OSI model.

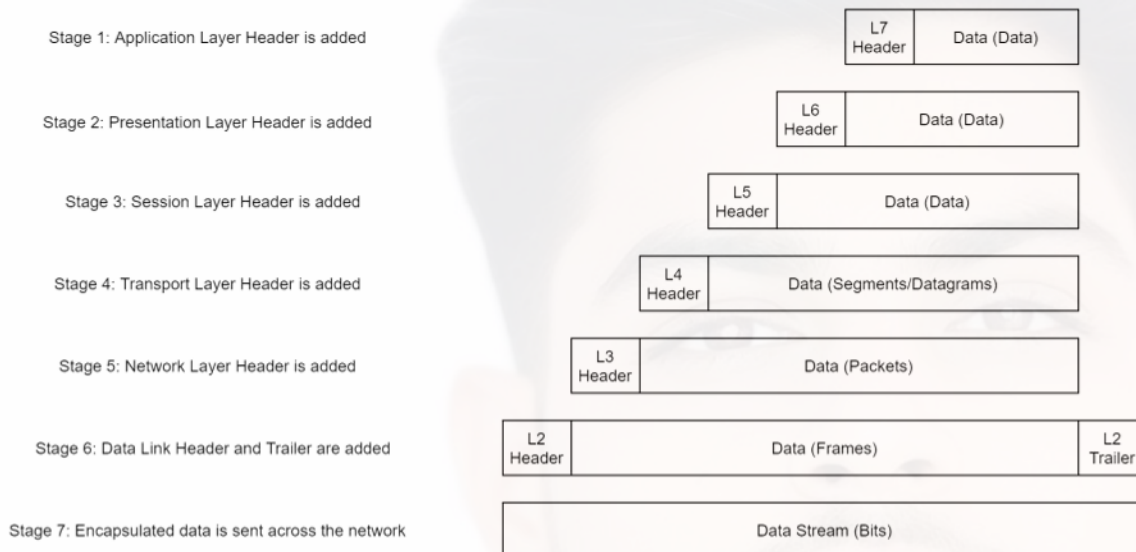
## Layer 2 - Data Link

The data link layer focuses on the physical addressing of the transmission. It receives a packet from the network layer (including the IP address for the remote computer) and adds in the physical **MAC** (Media Access Control) address of the receiving endpoint. Inside every network-enabled computer is a **Network Interface Card (NIC)** which comes with a unique MAC address to identify it.

MAC addresses are set by the manufacturer and literally burnt into the card; they can't be changed -- although they can be spoofed. When information is sent across a network, it's actually the physical address that is used to identify where exactly to send the information. Additionally, it's also the job of the data link layer to present the data in a format suitable for transmission.

## Layer 1 - Physical

This layer is one of the easiest layers to grasp. Put simply, this layer references the physical components of the hardware used in networking and is the lowest layer that you will find. Devices use electrical signals to transfer data between each other in a binary numbering system (1's and 0's).



## TCP/IP MODEL



The TCP/IP model is, in many ways, very similar to the OSI model. It's a few years older, and serves as the basis for real-world networking. The TCP/IP model consists of four layers: Application, Transport, Internet and Network Interface. Between them, these cover the same range of functions as the seven layers of the OSI Model.

| TCP/IP            |
|-------------------|
| Application       |
| Transport         |
| Internet          |
| Network Interface |

**Note:** Some recent sources split the TCP/IP model into five layers -- breaking the Network Interface layer into Data Link and Physical layers (as with the OSI model). This is accepted and well-known; however, it is not officially defined (unlike the original four layers which are defined in RFC1122). It's up to you which version you use -- both are generally considered valid.

You would be justified in asking why we bother with the OSI model if it's not actually used for anything in the real-world. The answer to that question is quite simply that the OSI model (due to being less condensed and more rigid than the TCP/IP model) tends to be easier for learning the initial theory of networking.

The two models match up something like this:

| OSI          | TCP/IP            |
|--------------|-------------------|
| Application  | Application       |
| Presentation |                   |
| Session      |                   |
| Transport    | Transport         |
| Network      | Internet          |
| Data Link    | Network Interface |
| Physical     |                   |

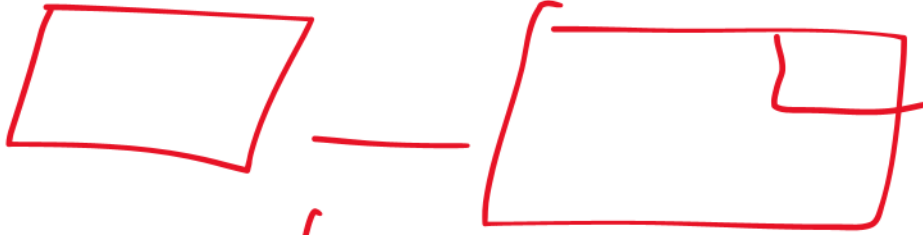
• TCP Flags:

| Flag | Name            | Function                                                                                                                                                              |
|------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SYN  | Synchronize     | Set during initial communication. Negotiating of parameters and sequence numbers (<br><b>SYN(Synchronised) - establish the connection between hosts and devices</b> ) |
| ACK  | Acknowledgement | Set as an acknowledgement to the SYN flag. Always set after initial SYN(<br><b>ACK(Acknowledgement) - means this verify if the data is reached there or not</b> )     |
| RST  | Reset           | Forces the termination of a connection (in both directions)(<br><b>RST(Reset) - resets a connection</b> )                                                             |
| FIN  | Finish          | Ordered close to communications(<br><b>FIN(finish) - No further transmission</b> )                                                                                    |
| PSH  | Push            | Forces the delivery of data without concern for buffering(<br><b>PSH(PUSH) - Send the buffer data immediately so the system can finish connection</b> )               |
| URG  | Urgent          | Data inside is being sent out of band. Example is cancelling a message( <b>URG(urgent) - data contained in the packet should be processed immediately</b> )           |

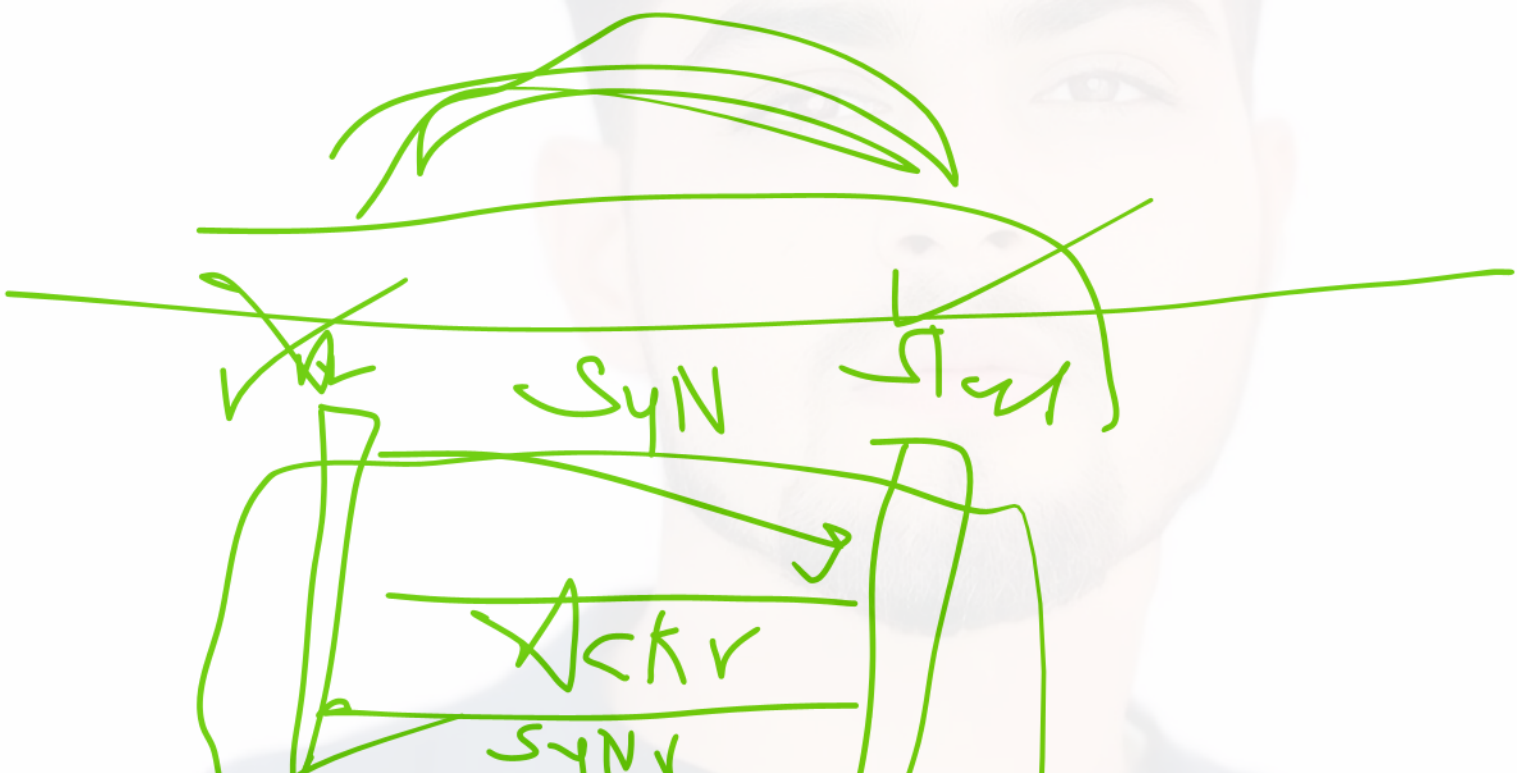
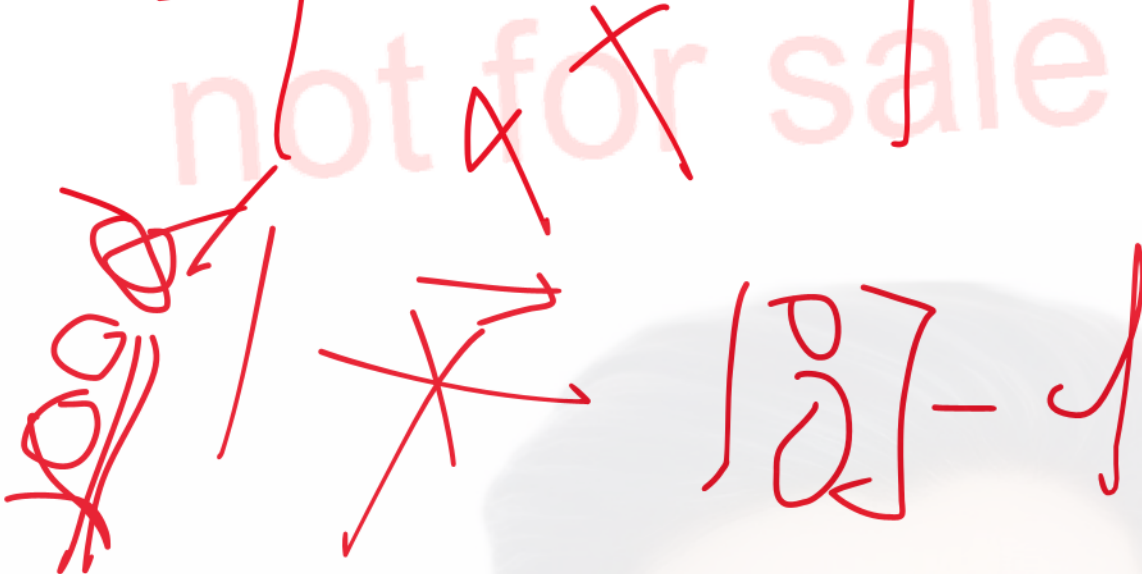
• History:

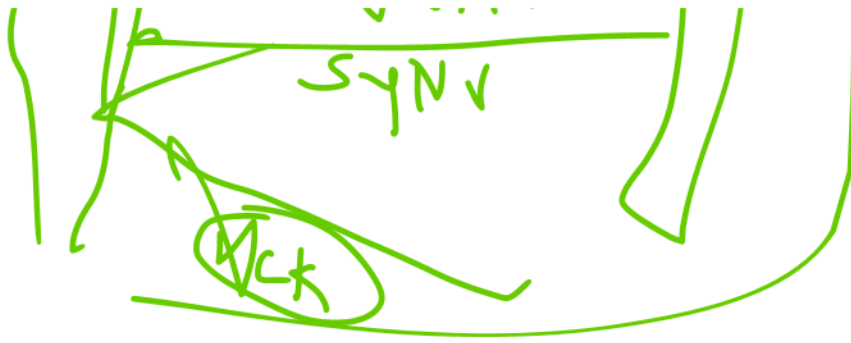
It's important to understand exactly why the TCP/IP and OSI models were originally created. To begin with there was no standardisation -- different manufacturers followed their own methodologies, and consequently systems made by different manufacturers were completely incompatible when it came to networking. The TCP/IP model was introduced by the American DoD in 1982 to provide a standard -- something for all of the different manufacturers to follow. This sorted out the inconsistency problems. Later the OSI model was also introduced by the International Organisation for Standardisation (ISO); however, it's mainly used as a more comprehensive guide for learning, as the TCP/IP model is still the standard upon which modern networking is based.

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